Red or Green, What Coloured Cat Food does Mi Prefer?

Vivian Dai

Question

Our beloved British Shorthair cat Mi loves to eat. We want to find out if Mi prefers red or green coloured cat food so he can enjoy his meals even more! Our hypothesis was that Mi has a preference for one of the colours. The null hypothesis is that Mi has no preference.

Method

Every day, in addition to his normal cat food, we gave Mi two bowls of cat food. One bowl had 10 pieces of red cat food and one bowl had 10 pieces of green cat food. Whichever food Mi started to eat first was counted as his preferred colour. He does not have to finish all the 10 pieces for his choice to count. The coloured cat food was prepared by mixing food colouring with his regular cat food. The position of the two bowls were switched around every day. We counted Mi's preference over 30 days, on trial for each day. Some days, Mi did not touch any food, so that day did not count.

Analysis

At the end of 30 days, we counted that Mi preferred red on 13 days and green on 17 days. To test our hypothesis that Mi has a preferred colour, we used the Chi-Squared Test of categories.

	Observed	**Expected**	$(O-E)^2$	$\frac{(O-E)^2}{E}$
Red	13	15	4	0.133
Green	17	15	4	0.133
Total				**0.266**

With one degree of freedom (red or green) the critical value for the Chi-Squared test at 95% confidence is 3.841. The calculated value was less than the critical value:

$$0.266 < 3.841$$

So we reject the hypothesis and accept the null hypothesis.

Conclusion

We reject our hypothesis that Mi has a preferred cat food colour. We can only assume Mi loves his food whether it is red or green!

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